

INTRODUCTION TO NLP

Natural language processing

Thanks to Dr. Ashraf Uddin Sir

Content

- ❑ What is NLP?
- ❑ Why is NLP?
- ❑ Tasks in NLP
- ❑ Challenges in NLP
- ❑ Terminologies

What is NLP?

Natural Language Processing (NLP) is a field of artificial intelligence that focuses on the interaction between computers and humans through natural language.

Its primary goal is to enable machines to **understand, interpret, and respond** to human language in a way that is both meaningful and useful.

Why Natural Language Processing?

- Bridging Human-Computer Communication
- Automation of Repetitive Language Tasks
- Efficient Information Extraction
- Enhances Data Accessibility and Usability
- Improves User Experience
- Applications in Various Industries
- Multilingual Capabilities

Why Natural Language Processing?

Amount of online textual data...

- 70 billion web-pages online (1.9 billion websites)
- 55 million Wikipedia articles

Growing at a fast pace

- 9000 tweets/second
- 3 million mail / second (60% spam)

Why Natural Language Processing?

JUL
2023

ESSENTIAL DIGITAL HEADLINES

OVERVIEW OF THE ADOPTION AND USE OF CONNECTED DEVICES AND SERVICES



GLOBAL OVERVIEW

TOTAL
POPULATION



we
are
social

8.05
BILLION

URBANISATION

57.2%

UNIQUE MOBILE
PHONE SUBSCRIBERS



Meltwater

5.56
BILLION

vs. POPULATION

69.1%

INDIVIDUALS USING
THE INTERNET



KEPIOS

5.19
BILLION

vs. POPULATION

64.5%

SOCIAL MEDIA
USER IDENTITIES



4.88
BILLION

vs. POPULATION

60.6%

10

SOURCES: UNITED NATIONS; GOVERNMENT BODIES; GSMA INTELLIGENCE; ITU; WORLD BANK; EUROSTAT; CNNIC; NIELSEN; CIA WORLD FACTBOOK; COMPANY ADVERTISING RESOURCES AND EARNINGS REPORTS; OECD; BETA RESEARCH CENTER; KEPIOS ANALYSIS. **ADVISORY:** SOCIAL MEDIA USER IDENTITIES MAY NOT REPRESENT UNIQUE INDIVIDUALS. **COMPARABILITY:** SIGNIFICANT REVISIONS TO SOURCE DATA, INCLUDING COMPREHENSIVE REVISIONS TO POPULATION DATA. FIGURES ARE NOT COMPARABLE WITH PREVIOUS REPORTS. ALL FIGURES USE THE LATEST AVAILABLE DATA, BUT SOME SOURCE DATA MAY NOT HAVE BEEN UPDATED IN THE PAST YEAR. SEE [NOTES ON DATA](#) FOR FULL DETAILS.

we
are
social

Meltwater

6

Why Natural Language Processing?

JUL
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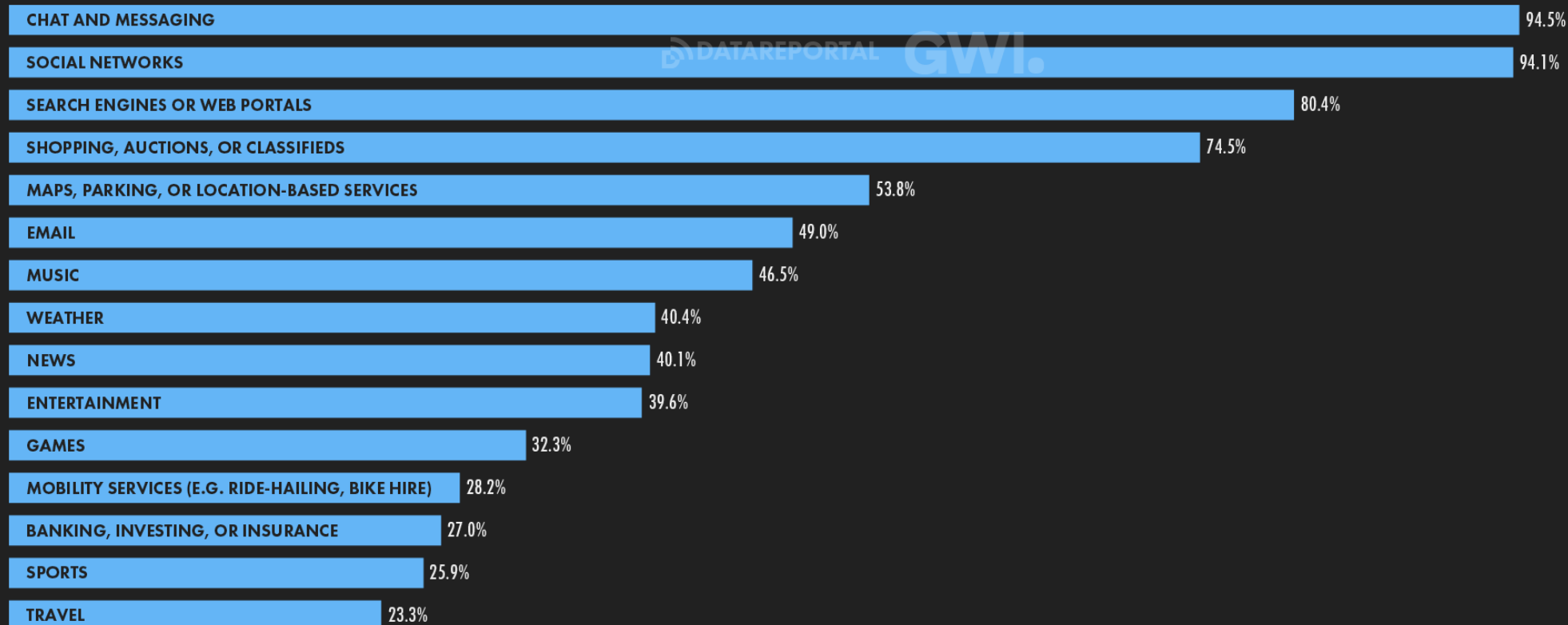
TOP TYPES OF WEBSITES VISITED AND APPS USED

PERCENTAGE OF INTERNET USERS AGED 16 TO 64 WHO HAVE VISITED OR USED EACH KIND OF DIGITAL PROPERTY IN THE PAST MONTH

NOTE: GWI HAVE REVISED THEIR METHODOLOGY. PLEASE READ THE IMPORTANT NOTES ON COMPARING DATA AT THE START OF THIS REPORT BEFORE COMPARING DATA ON THIS CHART WITH PREVIOUS REPORTS



GLOBAL OVERVIEW



NLP Tasks: Document Ranking





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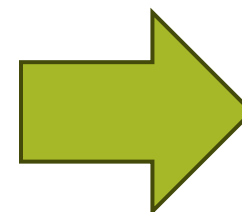
<https://www.cbsnews.com> › Sunday Morning

NLP Tasks: Document Classification

Saint Martin Island (Bengali: সেন্টমার্টিন দ্বীপ) is a small island (area only 3 km²) in the northeastern part of the Bay of Bengal, about 9 km south of the tip of the Cox's Bazar-Teknaf peninsula, and forming the southernmost part of Bangladesh. There is a small adjoining island that is separated at high tide, called Chera Dwip. It is about 8 kilometres (5 miles) west of the northwest coast of Myanmar, at the mouth of the Naf River.

History and description [\[edit\]](#)

Millennia ago, the island used to be an extension of the Teknaf peninsula, but at a later time some portion of this peninsula got submerged and thus the southernmost part of the aforementioned peninsula became an island, and was disconnected from the Bangladesh mainland. The first settlement started 250 years ago, in 18th century, by Arabian merchants who named the island 'Jazira'. During British occupation the island was named after the then Deputy Commissioner of Chittagong Mr. Martin as St. Martin Island.^[3] Likely because one or more of the Arabs were Saint and whose name could not be identified. Local names of the island are "Narikel jinjira"^[4] which means 'Coconut Island', and "Daruchini Dwip" which means "Cinnamon island". It is the only coral island in Bangladesh.



Sports

Travel

Politics

Economy

NLP Tasks: Named Entity Recognition

Identifying and classifying entities in a text into **predefined categories** such as names of persons, organizations, locations, dates, etc.

Example: Extracting "Google" as an organization and "California" as a location from the sentence: "Google is headquartered in California."

NLP Tasks: Part-of-Speech Tagging

Assigning **grammatical categories** (noun, verb, adjective, etc.) to each word in a sentence.

Example: In the sentence "The cat is sleeping," "The" is tagged as a determiner, "cat" as a noun, "is" as a verb, and "sleeping" as a verb.

NLP Tasks: Machine Translation

Automatically translating text from one language to another.

Example: Google Translate, which translates text between languages like English and Spanish.

NLP Tasks: Text Summarization

Generating a concise summary of a longer text while preserving its meaning.

Types:

Extractive Summarization: Extracts key sentences or phrases from the original text.

Abstractive Summarization: Generates new sentences that convey the core idea of the text.

Example: Summarizing a news article into a few key sentences.

NLP Tasks: Language Modeling

Predicting the next word in a sequence of words, given the previous words.

Example: Text autocompletion, such as predictive text on smartphones or autocomplete features in search engines.

NLP Tasks: Speech Recognition

Converting spoken language into text.

Example: Virtual assistants like Siri or Google Assistant transcribing spoken commands into text.

NLP Tasks: Text Generation

Generating human-like text based on a given prompt or context.

Example: GPT (Generative Pre-trained Transformer) models generating paragraphs of text based on an initial sentence or topic.

NLP Tasks: Question Answering (QA)

Answering questions posed in natural language, often based on a given context or knowledge base.

Types:

Open-domain QA: Can answer questions from any domain (e.g., Google search engine).

Closed-domain QA: Answers questions on a specific topic (e.g., a FAQ bot).

NLP Tasks: Coreference Resolution

Identifying when different words in a text refer to the same entity.

Example: In the sentence "**Alice** went to the store. **She** bought some apples," identifying that "She" refers to "Alice."

NLP Tasks: Sentiment Analysis

Analyzing the sentiment expressed in text, whether positive, negative, or neutral.

Example: Determining that the sentence "I love this movie!" expresses positive sentiment, while "I hated the book" expresses negative sentiment.

NLP Tasks: Topic Modeling

Automatically identifying topics or themes present in a collection of documents.

Example: Grouping news articles into topics like politics, sports, technology, and health

NLP Tasks: Dependency Parsing

Analyzing the grammatical structure of a sentence and identifying dependencies between words.

Example: Parsing the sentence "The dog chased the ball" to identify that "**dog**" is the subject and "**ball**" is the object of the verb "**chased.**"

NLP Tasks: Word Sense Disambiguation

Determining the correct meaning of a word in a given context when a word has multiple meanings.

Example: In the sentence "**I went to the bank to fish,**" identifying that "**bank**" refers to the side of a river, not a financial institution.

NLP Tasks: Dialogue Systems / Chatbots

Building systems that can engage in conversations with users, understand input, and generate appropriate responses.

Example: Virtual assistants like Alexa or customer service bots that help with queries.

NLP Tasks: Natural Language Inference

Determining whether a given hypothesis can be inferred from a premise, typically producing results like "entailment," "contradiction," or "neutral."

Example: Given the premise "**All dogs bark,**" the hypothesis "**My dog barks**" is an entailment, while "**My dog is silent**" is a contradiction.

NLP Tasks: Paraphrase Detection

Identifying whether two sentences have the same meaning or convey similar information.

Example: Checking if "The book was interesting" and "I found the book fascinating" are paraphrases.

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Challenges

1. Productivity
2. Ambiguous
3. Variability
4. Diversity
5. Sparsity

Why is Language Hard to Model?

1. Productivity

New words, senses, structure are introduced in languages all the time

Examples: staycation and social distance were added to the Oxford Dictionary in 2021

Why is Language Hard to Model?

2. Ambiguous

Most linguistic observations (speech, text) are open to several interpretations

We (Humans) disambiguate - i.e. find the correct interpretation - using
all kind of signals (linguistic and extra linguistic)

Why is Language Hard to Model?

3. Variation

- Phonetic (accent)
- Morphological, Lexical (spelling)
- Syntactic
- Semantic

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Essentially, the Variability of a language depends on:

- Social Context
- Geography
- Sociology
- Date
- Topic

Why is Language Hard to Model?

4. Diversity

About 7000 languages spoken in the world

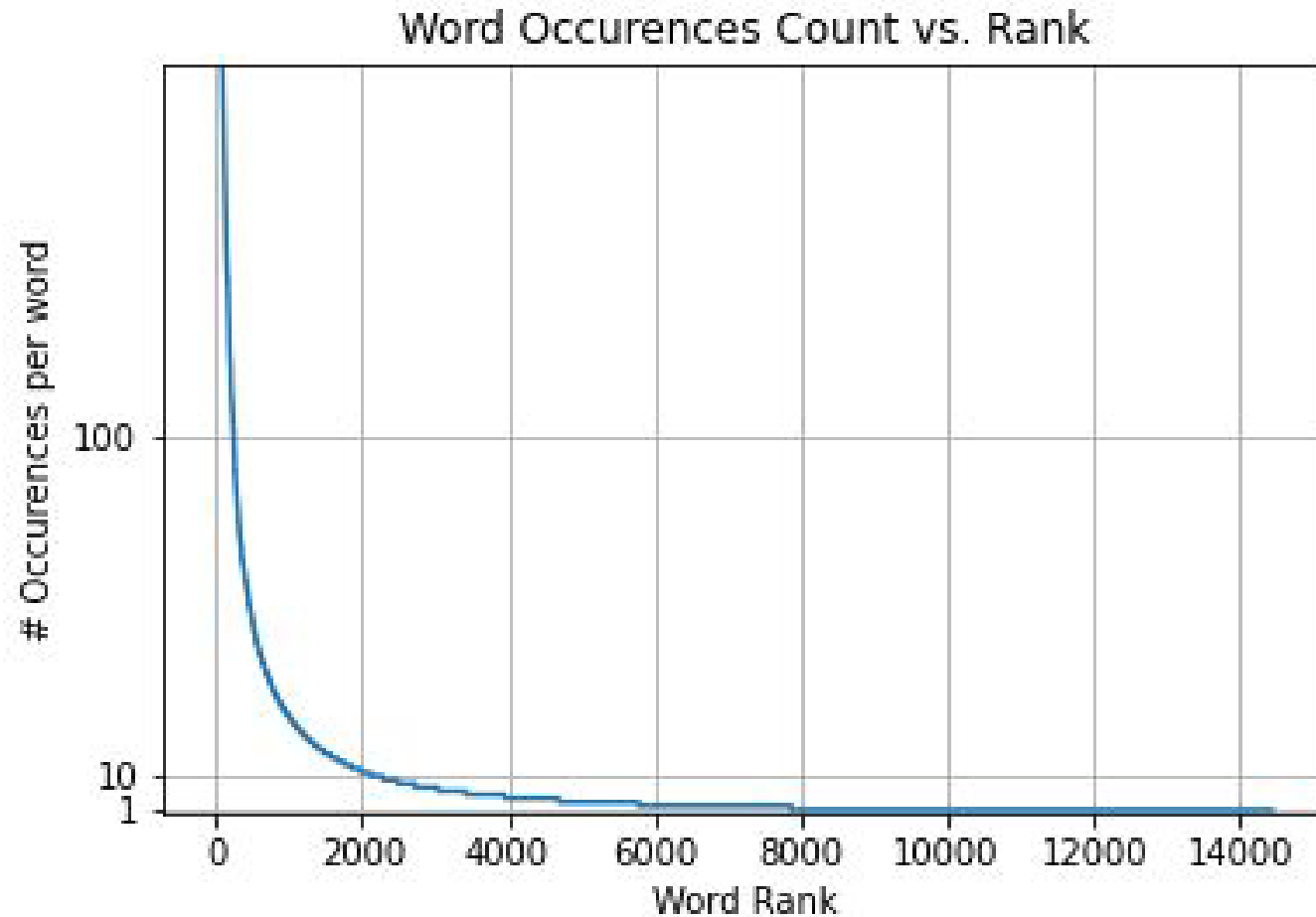
About 60% are found in the written form

Why is Language Hard to Model?

5. Sparsity

A large number of units are observed in language with very low frequency Very challenging for NLP

Why is Language Hard to Model?



In a large enough corpus, word distributions follows a Zipf Law

Word distribution in a corpus

Terminologies

A **token** is a sequence of characters (sometimes a word, sometimes several words, sometimes a sub-word...)

A **document** can be: A Sentence. A Paragraph. A sequence of characters

Terminologies

A **token** is the basic unit of discrete data, defined to be an item from a vocabulary indexed by $1, \dots, V$.

A **document** is a sequence of N words denoted by $d = (w_1, w_2, \dots, w_N)$, where w_N is the N -th word in the sequence.

A **corpus** (plural: **corpora**) is a collection of M documents denoted by $D = (d_1, d_2, \dots, d_M)$

Example: Wikipedia, All the articles of the NYT in 2021...

References

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2. "Natural Language Processing with Python" by Steven Bird, Ewan Klein, and Edward Loper
3. "Deep Learning for Natural Language Processing" by Jason Brownlee